Baptiste AUDUGÉ

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Engineering Student at CentraleSupélec | Applied AI

Seeking a 6-month applied science internship from February 2027 in London

Engineering student with a math & CS background, currently building **physics-informed deep learning models** for uranium mining at Orano. Designs new architectures from first principles to solve open industrial problems, from deriving governing equations to implementing and training them in PyTorch.

EXPERIENCE

Orano | *Global nuclear fuel cycle company* — Paris, France
Data Scientist Intern – Deep Learning & Generative AI — Jul 2026 – Jan 2027

- Asked to fix a **production AI model** planning well cleaning across uranium mines; traced its errors to an overlooked factor: **interactions between neighboring wells**.
- Designed a new **physics-informed model** capturing well-to-well interactions, deriving its governing equations from the geology literature.
- Built the data extraction (SQL, PI Web API) from the mine's monitoring systems; found **~30% of legacy data silently imputed**.
- Implementing the model in PyTorch: **Transformer with spatio-temporal cross-attention coupled to a Neural ODE**, trained on an NVIDIA DGX Spark on 10,000+ well time series spanning up to 20 years.

QSTNMRK | *Art Collective (Freelance)* — Paris, France
Lead Developer — Jan 2024 – Present

- Brought in to help on a campaign website; proposed and delivered a full rebuild, then became lead developer, owning code, hosting, payments and email workflows.
- *The Last Dollar* (2026, featured in **designboom**): live-streamed sculpture fed by an internet-connected ATM. Built it end-to-end, including a **24/7 streaming infrastructure** designed for flat cost regardless of audience.

Freelance Full-Stack Developer — Paris, France
Monaliza | *B2B2C financial products distributor, in partnership with* **Generali** — Jun 2022 – Sep 2024

- Built with the founders the real-time simulation engine (NestJS, React) that brokers use to sell the company's products; it became the core tool of investor presentations.

Lexem | *Wealth Management Firm* — Nov 2020 – Jul 2022

- Built a tax-impact simulator and client-intake platform with advisors; a key asset in the firm's sales strategy and partial acquisition.

EDUCATION

CentraleSupélec – Université Paris-Saclay | *Master's-level Engineering Degree (Diplôme d'Ingénieur)* — Paris, France
Coursework: Machine Learning, High-Performance Computing, Statistics, Signal Processing — 2024 – 2028 (expected)

Sorbonne University | *Double Bachelor's Degree in Mathematics & Computer Science* — Paris, France
Coursework: C Programming, Data Structures, Databases, Probability, Algebra — 2021 – 2024

LEADERSHIP & ACTIVITIES

Sono Barco CentraleSupélec | *Student Association* — Paris, France
Treasurer & Event Technician — 2025 – 2026

- Managed **€120k annual revenue**, tax filings (corporate tax, VAT) and the relationship with the accountant.
- Negotiated service contracts with HEC, ESSEC, Mines Paris; led event setups with **€300k+ of equipment** and **4500** attendees.

ACADEMIC PROJECTS

- **HPC Aircraft Drag Simulation:** CFD simulation parallelized with **MPI/OpenMP** and run on a **SLURM**-managed cluster.
- **IoT Anomaly Detection (Spark)**: anomaly detection and root-cause pipeline on large-scale IoT data from **Bouygues Telecom**.
- **Graphical Models in Finance**: conditional dependency detection across MSCI World assets using **Graphical Lasso**.

SKILLS & LANGUAGES

Languages: English (C1), French (native), Spanish (B1)
AI & Data Science: Deep learning (Transformers, Neural ODEs), physics-informed ML, time-series modeling, PyTorch, Scikit-learn
Software & Cloud: Python, C/C++, SQL, MPI, OpenMP, SLURM, Docker, AWS, Git, Linux, TypeScript (NestJS, React), Java
Client & Business: Stakeholder requirements gathering, contract negotiation, budgeting & accounting
Interests: Climbing, sound engineering & DJing, bass guitar, aeronautics (French Aviation Certificate, BIA)